
Heaven's Light Is Our Guide



RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

COURSE TITLE

Digital Techniques Sessional

Course No. ECE 2112

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REPORT TITLE

Simplification of Boolean Expressions Using Boolean Algebra and Karnaugh Map, and Verification Using Logic Circuit Simulation

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Problem: 01

Problem Definition: Simplification of the Boolean expression $F(A, B, C) = A'BC + AB'C + ABC' + ABC$ using Boolean algebra and the Karnaugh map, followed by verification of both the original and simplified circuits using logic simulation.

Objective:

To reduce a three-variable sum-of-minterms expression to its minimal sum-of-products form, to confirm the reduction using a Karnaugh map, and to verify that the original and simplified circuits are functionally identical by implementing both in Logisim.

Algebraic Simplification:

$$\begin{aligned}
 F(A, B, C) &= A'BC + AB'C + ABC' + ABC \\
 &= A'BC + ABC + AB'C + ABC + ABC' + ABC \\
 &= BC(A' + A) + AC(B' + B) + AB(C' + C) \\
 &= BC + AC + AB \\
 &= AB + AC + BC
 \end{aligned}$$

Truth Table:

A	B	C	$F = AB + AC + BC$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Table 1: Truth table for $F = AB + AC + BC$

Karnaugh Map:

$A \backslash BC$	00	01	11	10
0	0	0	1	0
1	0	1	1	1

Table 2: K-map for $F(A, B, C) = A'BC + AB'C + ABC' + ABC$

Three groups of two adjacent 1-cells are formed: the pair at $BC = 11$ (both rows) gives BC ; the pair at $A = 1, C = 1$ ($BC = 01, 11$) gives AC ; and the pair at $A = 1, B = 1$

($BC = 11, 10$) gives AB . Combining the three groups yields the minimized expression $F = AB + AC + BC$, matching the algebraic result.

Circuit Diagram:

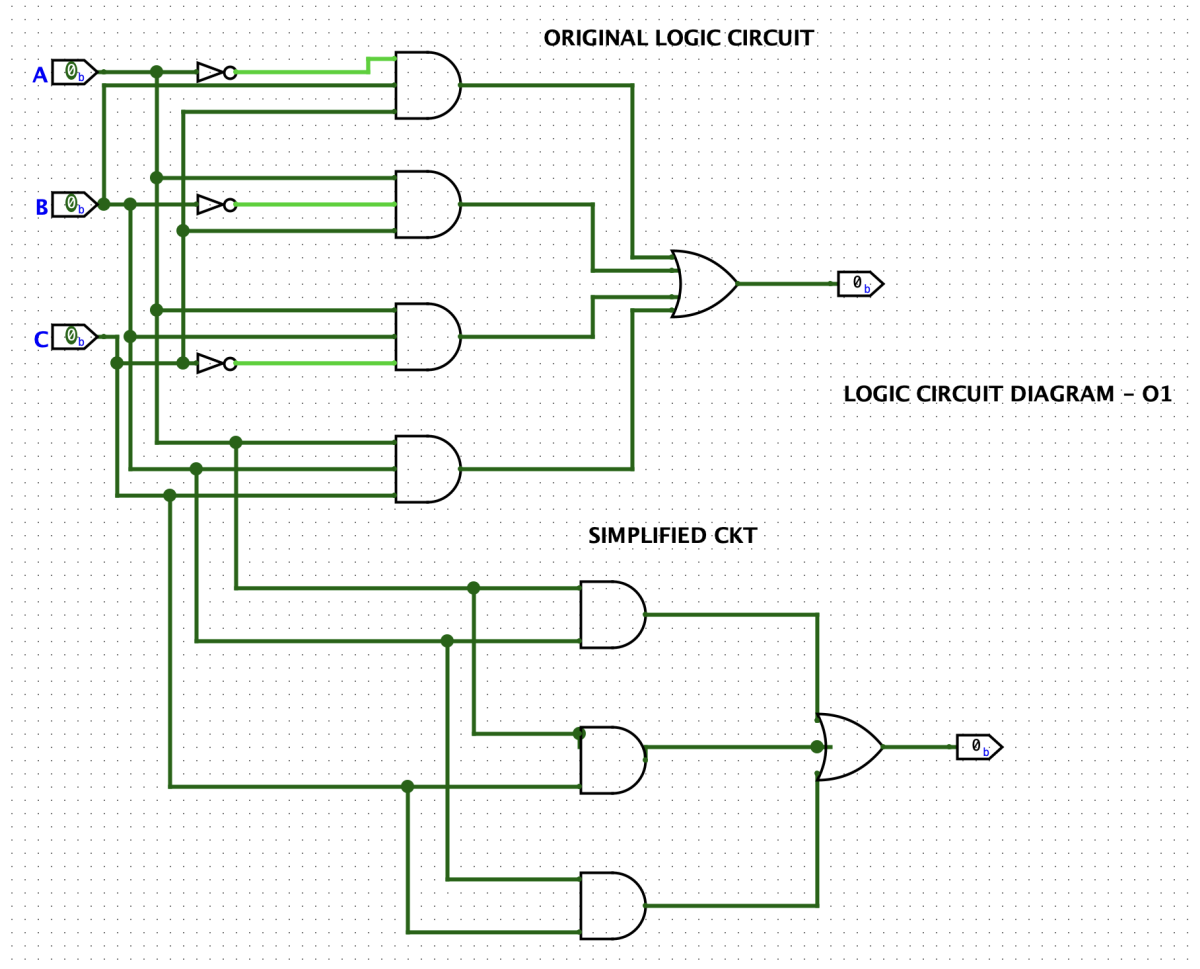


Figure 1: Original circuit (four 3-input AND gates plus a 3-input OR gate, realizing $A'BC + AB'C + ABC' + ABC$) and simplified circuit (three 2-input AND gates plus an OR gate, realizing $AB + AC + BC$), both simulated in Logisim

Discussion:

The original circuit implements the expression directly as given, requiring four 3-input AND gates (one per product term) and inverters to generate A' , B' , and C' where needed, followed by a 3-input OR gate. The simplified circuit realizes $AB + AC + BC$ using only three 2-input AND gates and one OR gate, with no inverters at all. Both circuits were driven with the same input combinations in Logisim, and the outputs matched for every case, confirming that the algebraic and Karnaugh-map simplifications are correct. The simplified circuit is noticeably more efficient, using fewer gates, fewer gate inputs, and no inverters.

Conclusion:

The expression $F = A'BC + AB'C + ABC' + ABC$ was successfully reduced to $F = AB + AC + BC$ using Boolean algebra and confirmed with a Karnaugh map. Logisim

simulation of both the original and simplified circuits produced identical outputs for all input combinations, verifying the correctness of the simplification.

Problem: 02

Problem Definition: Simplification of the Boolean expression $F(A, B, C) = A(A + B)(A + B + C)$ using Boolean algebra and the Karnaugh map, followed by verification of both the original and simplified circuits using logic simulation.

Objective:

To reduce a three-variable product-of-sums-type expression to its simplest form, to confirm the reduction using a Karnaugh map, and to verify the equivalence of the original and simplified circuits in Logisim.

Algebraic Simplification:

$$\begin{aligned}
 F(A, B, C) &= A(A + B)(A + B + C) \\
 &= (A + AB)(A + B + C) \\
 &= A(A + B + C) + AB(A + B + C) \\
 &= A + AB + AC + AB + AB + ABC \\
 &= A + AB + AC + ABC \\
 &= A(1 + B + C + BC) \\
 &= A
 \end{aligned}$$

Truth Table:

A	B	C	$F = A$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Table 3: Truth table for $F = A$

Karnaugh Map:

All four cells in the $A = 1$ row are 1, while the entire $A = 0$ row is 0. This forms a single group of four adjacent cells covering the whole row, which eliminates both B and C and leaves only $F = A$, exactly matching the algebraic simplification.

$A \backslash BC$	00	01	11	10
0	0	0	0	0
1	1	1	1	1

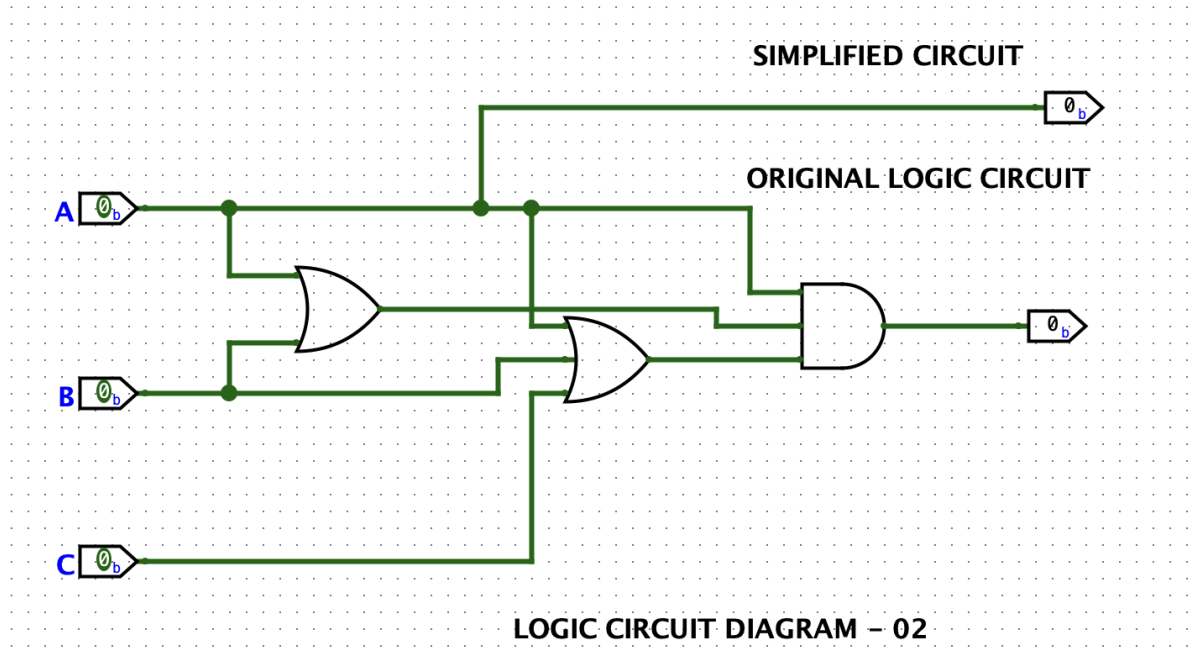
Table 4: K-map for $F(A, B, C) = A(A + B)(A + B + C)$ 

Figure 2: Original circuit (two OR gates realizing $A + B$ and $A + B + C$, combined with A through an AND gate) and simplified circuit (a direct connection representing $F = A$), simulated in Logisim

Circuit Diagram:

Discussion:

The original circuit forms $A + B$ with one OR gate, extends it to $A + B + C$ with a second OR gate, and ANDs both results with A itself. Because A is common to every term, all higher-order terms are absorbed and the entire network collapses to the single literal A , so the simplified circuit is simply a wire from input A to the output, with no logic gates required. This is an example of the absorption law $A + AB = A$ applied twice. Testing both circuits in Logisim for all combinations of A , B , and C showed that the output always equals A regardless of B and C , confirming the simplification.

Conclusion:

The expression $F = A(A + B)(A + B + C)$ was shown to reduce to $F = A$ through repeated application of the absorption law, and this result was confirmed by both the Karnaugh map and Logisim simulation, which showed that B and C have no effect whatsoever on the output.

Problem: 03

Problem Definition: Simplification of the Boolean expression $F(A, B, C) = (A + (BC)')'(AB + ABC)$ using Boolean algebra and the Karnaugh map, followed by verification of both the original and simplified circuits using logic simulation.

Objective:

To reduce a mixed AND-OR-complement expression using De Morgan's theorem and Boolean algebra, to confirm the result with a Karnaugh map, and to verify using Logisim that the simplified circuit behaves as a constant.

Algebraic Simplification:

$$\begin{aligned}
 F(A, B, C) &= (A + (BC)')'(AB + ABC) \\
 &= A' \cdot BC \cdot (AB + ABC) \quad (\text{De Morgan's theorem}) \\
 &= A'BC \cdot AB(1 + C) \\
 &= A'BC \cdot AB \\
 &= (AA')BCB \\
 &= 0
 \end{aligned}$$

Truth Table:

A	B	C	$F = 0$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

Table 5: Truth table for $F = 0$

Karnaugh Map:

$A \backslash BC$	00	01	11	10
0	0	0	0	0
1	0	0	0	0

Table 6: K-map for $F(A, B, C) = (A + (BC)')'(AB + ABC)$

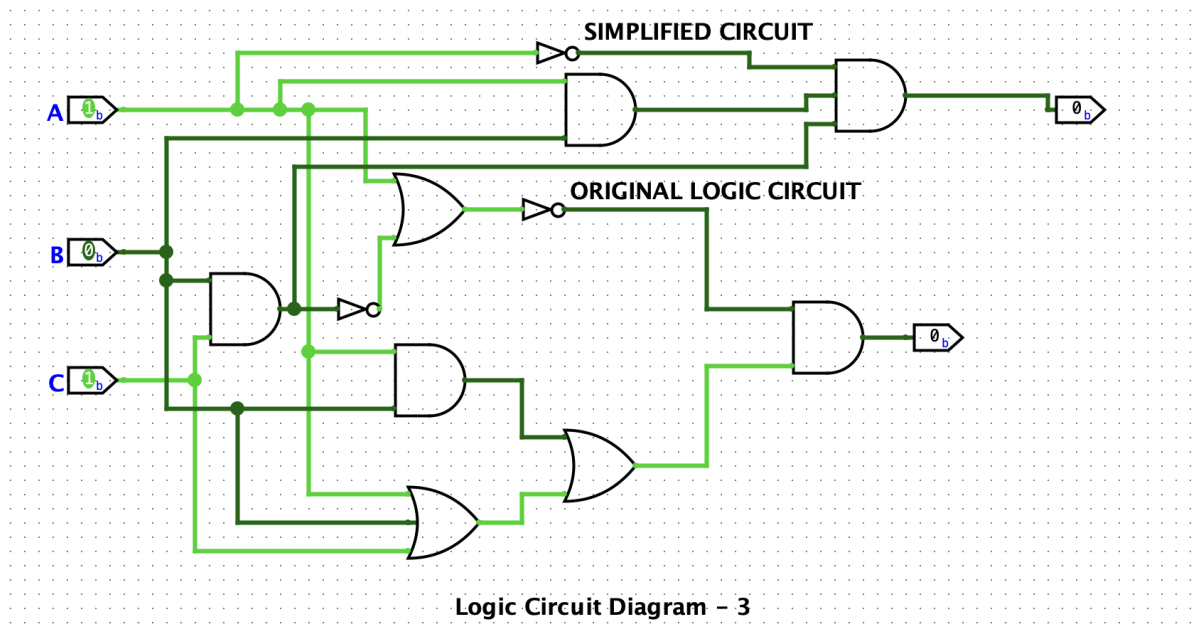


Figure 3: Original circuit realizing $(A + (BC)')'(AB + ABC')$ using NAND-type and OR/AND stages, and simplified circuit showing the output permanently at logic 0, simulated in Logisim

Every cell of the map is 0, so there are no 1-cells to group. A Karnaugh map with no populated cells represents the constant function $F = 0$, confirming the algebraic result.

Circuit Diagram:

Discussion:

Applying De Morgan's theorem to $(A + (BC)')'$ converts it to $A' \cdot BC$, after which the second factor $AB + ABC$ simplifies to AB by absorption ($AB + ABC = AB(1 + C) = AB$). Multiplying $A'BC$ by AB introduces the term $A \cdot A'$, which is always 0, forcing the entire expression to 0 regardless of the values of A , B , and C . This was confirmed in Logisim by cycling through all eight input combinations on the original circuit and observing that the output never leaves logic 0, exactly as predicted by the all-zero Karnaugh map.

Conclusion:

The expression $F = (A + (BC)')'(AB + ABC)$ was shown to be identically zero for all input combinations. This was verified algebraically, confirmed by an empty Karnaugh map, and validated by Logisim simulation, which showed a constant logic-0 output regardless of A , B , and C .

Problem: 04

Problem Definition: Simplification of the Boolean expression $F(A, B) = (B'(A + B) + (A + B)(A + B'))B'$ using Boolean algebra and the Karnaugh map, followed by verification of both the original and simplified circuits using logic simulation.

Objective:

To reduce a two-variable expression built from nested OR and AND terms to its minimal form, to confirm the reduction using a Karnaugh map, and to verify the equivalence of the original and simplified circuits in Logisim.

Algebraic Simplification:

$$\begin{aligned}
 F(A, B) &= (B'(A + B) + (A + B)(A + B'))B' \\
 &= (AB' + BB' + AA + AB' + AB' + BB')B' \\
 &= (A + AB' + AB' + AB')B' \\
 &= A(1 + B' + B' + B')B' \\
 &= AB'
 \end{aligned}$$

Truth Table:

A	B	$F = AB'$
0	0	0
0	1	0
1	0	1
1	1	0

Table 7: Truth table for $F = AB'$

Karnaugh Map:

$A \backslash B$	0	1
0	0	0
1	1	0

Table 8: K-map for $F(A, B) = (B'(A + B) + (A + B)(A + B'))B'$

Only a single cell, at $A = 1, B = 0$, is populated. A lone 1-cell cannot be combined with any adjacent cell, so it is read directly as the product of its coordinate literals, giving $F = AB'$, matching the algebraic result.

Circuit Diagram:

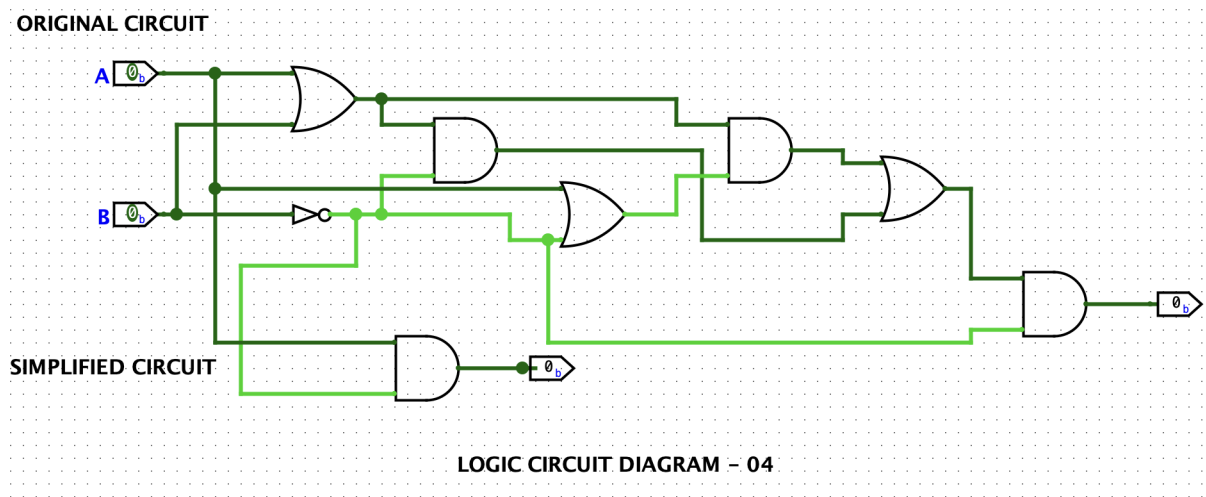


Figure 4: Original circuit realizing $(B'(A+B) + (A+B)(A+B'))B'$ using OR, AND, and NOT stages, and simplified circuit reduced to a single AND gate realizing AB' , simulated in Logisim

Discussion:

Expanding the original expression term by term produces several instances of AB' along with the terms A , BB' , and AA ; since $BB' = 0$ and $AA = A$, and every surviving term is either A or AB' , factoring out the common literal B' across the whole expression collapses it to the single term AB' . Consequently, the four-gate original network (built from OR, AND, and NOT gates as shown in the Logisim schematic) is functionally equivalent to a single 2-input AND gate with B inverted. Verifying both circuits in Logisim for all four input combinations of A and B confirmed that the output is 1 only when $A = 1$ and $B = 0$, matching $F = AB'$ in every case.

Conclusion:

The expression $F = (B'(A+B) + (A+B)(A+B'))B'$ was successfully reduced to $F = AB'$ using Boolean algebra, confirmed with a Karnaugh map, and validated by Logisim simulation, which showed identical outputs for the original and simplified circuits across all input combinations.

References

1. M. M. Mano, *Digital Design*, 5th ed. Upper Saddle River, N.J.: Pearson, 2013.
2. R. J. Tocci, N. S. Widmer, and G. L. Moss, *Digital Systems: Principles and Applications*, 11th ed. Upper Saddle River, N.J.: Pearson, 2011.